MVUE : 1. CRLB 2. Linear Model 3. Sufficient

CRLB $Var(\hat{\theta}) \geq I(\theta)^{-1}$

Effective 取等 $s(x;\theta) = \frac{\partial}{\partial\theta}\ln f(x;\theta) = I(\theta)(g(x)-\theta)$ $\hat{\theta}_{MVU} = g(x)$, $Var(\hat{\theta}) = I(\theta)^{-1}$

Fisher Information $I(\theta) = -E[\frac{\partial^2}{\partial\theta^2}\ln f(x|\theta)]$ when $E[s(x;\theta)] = 0$

Linear Model $x = H\theta + w$ WGN. $s(x;\theta) = \frac{1}{\sigma^2}(H^Tx - H^TH\theta)$

$\hat{\theta}_{MVU} = (H^TH)^{-1}H^Tx$, $Var(\hat{\theta}) = \frac{1}{\sigma^2}(H^TH)^{-1}$

$\alpha = g(\theta)$

$Var(\hat{\alpha}) = [g'(\theta)]^2 I^{-1}(\theta)$

$C(\hat{\alpha}) = (\nabla g)^T I^{-1}(\theta)(\nabla g)$

Sufficient Statistics $p(x|T;\theta) = p(x|T)$

1) Neyman-Fisher factorization $p(x;\theta) = g(T(x),\theta)h(x)$ Find $T(x)$

$x \sim N(A, \sigma^2)$, $T_1(x) = \sum x[n]$, $T_2(x) = \sum x^2[n]$

2) $E(T(x))$, Find $T'(x)$ such that $E(T(x)) = \theta$

$E(T_1(x)) = NA$, $E(T_2(x)) = N(A^2+\sigma^2)$, $\hat{A}_{MVU} = \frac{1}{N}T_1(x) = \bar{x}$

BLUE $x = \theta s + w$ Let $\hat{\theta}_{BLUE} = a^Tx$ $\min a^TC_xa$ s.t. $a^Ts = 1$

$\hat{\theta}_{BLUE} = \frac{s^TC_x^{-1}x}{s^TC_x^{-1}s}$ $\hat{a}_{BLUE} = \frac{s^TC_x^{-1}}{s^TC_x^{-1}s}$ $MSE = Var(\hat{\theta}_{BLUE}) = \frac{1}{s^TC_x^{-1}s}$

$\hat{\theta}_{BLUE} = (H^TC_x^{-1}H)^{-1}H^TC_x^{-1}x$ $MSE = (H^TC_x^{-1}H)^{-1}$

ML $\hat{\theta}_{ML} = \arg\max p(x;\theta)$ asymptotic properties $\hat{\theta}_{ML} \sim N(\theta, I^{-1}(\theta))$

$N \to \infty$

$\alpha = \varphi(\theta)$ 则 $\hat{\alpha}_{ML} = \varphi(\hat{\theta}_{ML})$

MMSE $\hat{\theta}_{MMSE} = \arg\min BMSE = \arg\min E_\theta[L(\theta,\hat{\theta})|x]$

$BMSE = E_\theta E_x[(\theta-\hat{\theta})^2] = \int_\theta\int_x(\theta-\hat{\theta})^2 p(x,\theta)dxd\theta$ $p(x,\theta) = p(x)p(\theta|x)$

$= \int_x[\underbrace{\int_\theta(\theta-\hat{\theta})^2 p(\theta|x)d\theta}_{\text{给定 } x\text{，变化 } \hat{\theta}\text{，最小化}}]p(x)dx$ $\hat{\theta}_{MMSE} = \arg\min E_\theta[(\theta-\hat{\theta})^2|x]$

$\frac{\partial}{\partial\hat{\theta}}\int_\theta(\theta-\hat{\theta})^2 p(\theta|x)d\theta = 0$

$\hat{\theta}_{MMSE} = E_\theta(\theta|x) = \int_\theta\theta p(\theta|x)d\theta \longleftarrow p(\theta|x) = \frac{p(\theta)p(x|\theta)}{p(x)} = \frac{p(\theta)p(x|\theta)}{\int_\theta p(\theta)p(x|\theta)d\theta}$

$BMSE(\hat{\theta}_{MMSE}) = E_x(Var(\theta|x)) = \int_x Var(\theta|x)p(x)dx$

$= \int_x[\int_\theta(\theta-E_\theta(\theta|x))^2 p(\theta|x)d\theta]p(x)dx$

$\alpha = A\theta + b$,

则 $\alpha_{MMSE} = A\theta_{MMSE} + b$

LMMSE $\hat{\theta} = a^Tx$ $\min BMSE(a^Tx)$

x, θ joint Gaussian <=> LMMSE is MMSE

Gaussian $E(Y|X) = E(Y) + C_{xy}C_{xx}^{-1}[x - E(x)]$ $Cov(Y|X) = C_{yy} - C_{xy}C_{xx}^{-1}C_{xy}$

$x = H\theta + w$ $\theta \sim N(\mu_\theta, C_\theta)$, $w \sim N(0, C_w)$

$\hat{\theta}_{MMSE} = \mu_\theta + C_\theta H^T(HC_\theta H^T + C_w)^{-1}(x - H\mu_\theta)$ $\hat{\theta} = C_\theta(C_\theta + C_w)^{-1}x$

$BMSE(\hat{\theta}_{MMSE}) = C_\theta - C_\theta H^T(HC_\theta H^T + C_w)^{-1}HC_\theta = (C_\theta^{-1} + HC_w^{-1}H^T)^{-1}$

$\hat{\theta}_{MAP} = \arg\max p(\theta|x) = \frac{\partial}{\partial\theta}\ln p(x|\theta) + \frac{\partial}{\partial\theta}\ln p(\theta)$ 的根 $\hat{\theta}_{MMAE}$: $\int_\theta^\infty p(\theta|x)d\theta = \frac{1}{2}$

$p(x,y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}}\exp\{-\frac{1}{2(1-\rho^2)}[\frac{(x-\mu_1)^2}{\sigma_1^2} + \frac{(y-\mu_2)^2}{\sigma_2^2} - \frac{2\rho(x-\mu_1)(y-\mu_2)}{\sigma_1\sigma_2}]\}$

$p(x;\mu,C) = \frac{1}{(2\pi)^{N/2}|C|^{1/2}}\exp\{-\frac{1}{2}(x-H\theta)^TC^{-1}(x-H\theta)\}$

$(A+BCD)^{-1} = A^{-1} - A^{-1}B(DA^{-1}B + C^{-1})^{-1}DA^{-1}$

$(A+uu^H)^{-1} = A^{-1} - \frac{A^{-1}uu^HA^{-1}}{1+u^HAu}$

LS $\hat{\theta}_{LS} = \arg\min \sum (x[n] - \theta[n])^2$

Linear Model $x = H\theta + w$, $\hat{\theta}_{LS} = (H^TH)^{-1}H^Tx$. $J_{min} = x^T P_H^{\perp} x$ $P_H = H(H^TH)^{-1}H^T$

Weighted LS $\hat{\theta}_{LS} = (H^TWH)^{-1}HW^Tx$. $J_{min} = x^T[W - WH(H^TWH)^{-1}H^TW]x$

DC in WGN

$H_0: x[n] = w[n]$

$H_1: x[n] = A + w[n]$

NP Rule

$$L(x) = \frac{P(x;H_1)}{P(x;H_0)} > \gamma$$

$$T(x) := \bar{x} > \gamma' = \frac{\sigma^2}{NA}\ln\gamma + \frac{1}{2}A$$

$$T \sim \begin{cases} N(0, \frac{\sigma^2}{N}), & H_0 \\ N(A, \frac{\sigma^2}{N}), & H_1 \end{cases}$$

$$P_{FA} = P(T > \gamma'; H_0) = Q\left(\frac{\gamma'}{\sigma/\sqrt{N}}\right) = \alpha$$

$$P_D = P(T > \gamma'; H_1) = Q\left(\frac{\gamma' - A}{\sigma/\sqrt{N}}\right)$$

Min P_e

$i = \arg\max_{i=0,1} P(x|H_i)$

$$L(x) = \frac{P(x;H_1)}{P(x;H_0)} > \gamma = \frac{P(H_0)}{P(H_1)} = 1$$

$$T(x) = \bar{x} > \frac{1}{2}\bar{A}$$

$$P_e = \frac{1}{2}[P(H_0|H_1) + P(H_1|H_0)]$$

$$= \frac{1}{2}[P(\bar{x} < \tfrac{1}{2}A|H_1) + P(\bar{x} > \tfrac{1}{2}A|H_0)]$$

$$T \sim \begin{cases} N(0, \frac{\sigma^2}{N}), & H_0 \\ N(A, \frac{\sigma^2}{N}), & H_1 \end{cases}$$

$$= P(\bar{x} > \tfrac{1}{2}A|H_0) = P\left(\frac{\bar{x}}{\sigma/\sqrt{N}} > \frac{\frac{1}{2}A}{\sigma/\sqrt{N}} \,\Big|\, H_0\right)$$

$$= Q\left(\sqrt{\frac{NA^2}{4\sigma^2}}\right)$$

Bayes

$i = \arg\max_{i=0,1} P(x|H_i)P(H_i)$

Random Detection

$H_0: x[n] = w[n]$ $s[n]$ 未知且随机

$H_1: x[n] = s[n] + w[n]$

当 $s \sim N(0, \sigma_s^2 I)$, $w \sim N(0, \sigma^2 I)$

$$L(x) = \frac{P(x;H_1)}{P(x;H_0)} > \gamma \quad T(x) := \sum x^2[n] > \gamma'$$

$$\begin{cases} T/\sigma^2 \sim \chi_N^2, & H_0 \\ T/\sigma^2 + \sigma_s^2 \sim \chi_N^2, & H_1 \end{cases}$$

$$P_{FA} = P(T > \gamma'; H_0) = Q\left(\frac{\gamma'}{\sigma^2}\right) = \alpha$$

$$P_D = P(T > \gamma'; H_1) = Q\left(\frac{\gamma'}{\sigma^2 + \sigma_s^2}\right)$$

Matched Filter

$H_0: x[n] = w[n]$

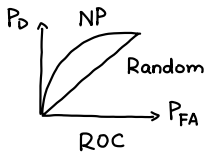
$H_1: x[n] = s[n] + w[n]$ $s[n]$ 未知且确定

NP Rule

$$L(x) = \frac{P(x;H_1)}{P(x;H_0)} > \gamma$$

$$T(x) := x^Ts > \gamma' = \sigma^2\ln\gamma + \frac{1}{2}\sum_{n=0}^{N-1} s^2[n]$$

$$T \sim \begin{cases} N(0, \sigma^2\mathcal{E}), & H_0 \\ N(\mathcal{E}, \sigma^2\mathcal{E}), & H_1 \end{cases} \quad \mathcal{E} = \|s\|^2$$



$$P_{FA} = P(T > \gamma'; H_0) = Q\left(\frac{\gamma'}{\sqrt{\sigma^2\mathcal{E}}}\right) = \alpha$$

$$P_D = P(T > \gamma'; H_1) = Q\left(\frac{\gamma' - \mathcal{E}}{\sqrt{\sigma^2\mathcal{E}}}\right) = Q\left(Q^{-1}(\alpha) - \sqrt{\frac{\mathcal{E}}{\sigma^2}}\right)$$

max SNR $x^Ts > \gamma'$ $\eta = \frac{1}{\sigma^2}\frac{(h^Ts)^2}{h^Th}$

whenever w is WGN

$w \sim N(0, C_w)$

$$T(x) := x^TC_w^{-1}s > \gamma'$$

$$P_D = Q(Q^{-1}(\alpha) - \sqrt{d^2}) \quad d^2 = s^TC^{-1}s$$

已知 $\mathcal{E}$, max P_D $\Rightarrow$ $s = C^{-1}$ 最大 λ 的特征向量

Min P_e

$$i = \arg\max_i P(x|H_i) = \arg\min \|x - s_i\|^2$$

$$T_i(x) = x^Ts_i - \frac{1}{2}\mathcal{E}_i$$

当二分类 $T(x) = T_1(x) - T_0(x)$

$$P_e = P(T(x) > 0|H_0) = Q\left(\frac{1}{2}\sqrt{\frac{\|s_0 - s_1\|_2^2}{\sigma^2}}\right)$$

已知 $\bar{\mathcal{E}} = \frac{1}{2}(\mathcal{E}_0 + \mathcal{E}_1)$, $\|s_0 - s_1\|_2^2 = 2\bar{\mathcal{E}}(1-\rho)$

$$\rho = \frac{s_1^Ts_0}{\frac{1}{2}(s_1^Ts_1 + s_0^Ts_0)} \in [-1, 1] \quad P_e = Q\left(\sqrt{\frac{\bar{\mathcal{E}}(1-\rho)}{2\sigma^2}}\right)$$

$$\geq Q(\sqrt{\bar{\mathcal{E}}/\sigma^2})$$

当 $s \sim N(0, C_s)$, $w \sim N(0, C_w)$

$$T(x) := x^T\hat{s} = x^TC_s(C_s + C_w)^{-1}x > \gamma''$$

当 Linear Model $x = H\theta + w$ ($\hat{s}$ is LMMSE)

Let $s = H\theta$ $C_s = HC_\theta H^T$